22 Principles Based on Social Cues in Multimedia Learning

Personalization, Voice, Image, and Embodiment Principles

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WHAT ARE THE PRINCIPLES BASED ON SOCIAL CUES IN MULTIMEDIA LEARNING?

Multimedia research has primarily focused on ways to reduce extraneous cognitive processing – processing irrelevant to the instructional goal (see Chapter 14) – and manage essential processing – processing related to representing the essential material (see Chapter 19). However, a third instructional goal derived from the cognitive theory of multimedia learning (described in Chapter 5) is that multimedia lessons should foster generative processing – cognitive processing related to making sense of the learning material. Generative processing depends on students' motivation to learn. If students do not believe learning is worth the effort, they will not engage in the cognitive processes necessary for deep learning, such as organizing the material into a coherent structure and integrating it with their existing knowledge. This chapter explores how incorporating social cues within multimedia lessons – including aspects of the instructor's speech, presence, and body movements – can foster student motivation and meaningful learning.

One challenge of many multimedia learning environments, including illustrated textbooks, narrated animations, and instructional videos, is that they may feel isolating or impersonal to students. Without a feeling of social support – that is, a feeling that others have an interest in their learning – students will be less inclined to invest effort toward actively engaging with the material. The rationale for integrating social cues within multimedia lessons is to create a feeling of social connection even when a human instructor is not physically present (Nass & Brave, 2005; Reeves & Nass, 1996). Common social cues include the instructor's (or on-screen agent's) style of language (e.g., conversational or formal), voice (e.g., human or machine-generated), presence (e.g., visible or not visible), and level of embodiment (e.g., use of gesture and eye-gaze cues). According to social agency theory (Mayer & DaPra, 2012; Mayer, Fennell, Farmer, & Campbell, 2004), the presence of social cues activates a social response in the learner, which motivates them to engage in generative processing, resulting in meaningful learning outcomes (such as higher transfer test performance), as illustrated in Figure 22.1. The key component of social agency theory is activation of a social response, in which learners feel a sense of social presence, or partnership, between them and the instructor. When learners feel as if the instructor is investing effort to help them learn, they are more likely to uphold their end of the partnership and invest effort to make sense of the instructional message (Grice, 1975).

This chapter explores four types of social cues for fostering generative processing in multimedia learning: personalization, voice, image, and embodiment. The *personalization principle* is that people learn better when multimedia lessons are presented in a conversational or polite style, rather than a formal or direct style. The *voice principle* is that people learn better when multimedia lessons are presented in a human voice rather than computer-generated voice. *The image principle* is that people do not necessarily learn better when the image of the instructor or a pedagogical agent is visible on the screen. The *embodiment principle* is that people learn better when the instructor or onscreen pedagogical agents engages in human-like movement, such as using gestures, eye-contact, or facial expressions.

WHAT ARE EXAMPLES OF PRINCIPLES BASED ON SOCIAL CUES IN MULTIMEDIA LEARNING?

Personalization Cues

Can subtle changes to the style of an instructional message lead to better learning? In a series of five experiments by Moreno and Mayer (2000), college students learned from a multimedia lesson on lightning formation or a computer game on environmental science. Instructional messages were either presented in a conversational style (from the first- or second-person point of view) or a formal style (from the third-person point of view). For example, consider the following message from the environmental

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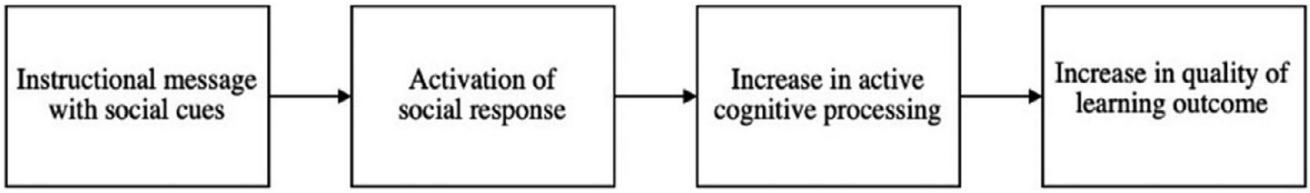

Figure 22.1 How social cues affect meaningful learning

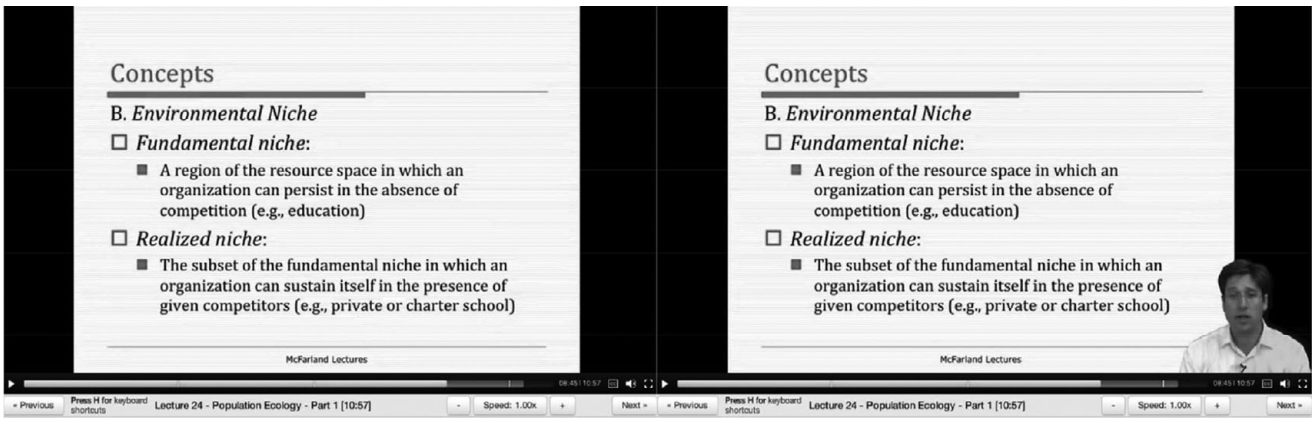

Figure 22.2 Video lesson without (left) and with (right) the instructor's image (from Kizilcec, Bailenson, and Gomez, 2015)

science game, written in formal style: "In very rainy environments plant leaves have to be flexible so that they are not damaged by the rainfall. What really matters for the rain is the choice between thick and thin leaves." To make this message more personalized, it could be modified to explicitly reference the learner, for example: "This is a very rainy environment, and the leaves of your plant have to be flexible so they're not damaged by the rainfall. What really matters for the rain is your choice between thick leaves and thin leaves. Which do you think would be more flexible?" Across all five experiments, students in the conversational style group significantly outperformed students in the formal style group on a subsequent transfer test. In a follow-up study, Moreno and Mayer (2004) replicated the personalization effect in a virtual learning environment.

Voice Cues

A related issue is whether learning from more natural or familiar instructional messages results in deeper learning. Mayer, Sobko, and Mautone (2003) explored how different types of voice cues affect learning from a narrated animation about lightning formation. Across two experiments, the narration was presented as an appealing human voice in the learners' native accent or a machine synthesized voice. Students who learned from a native human voice outperformed the other condition on a subsequent transfer test. These results demonstrate the advantage of using a human over machine voice. According to social agency theory, more natural or familiar voice cues create a stronger feeling of human-to-human communication during learning, motivating students to actively engage with the learning material.

Image Cues

Should the instructor's image be visible during multimedia lessons? A large field study by Kizilcec, Bailenson, and Gomez (2015) explored students' preferences for learning from video lessons with and without the instructor's image visible, as shown in Figure 22.2. Students reported mixed preferences: those who chose to view the instructor's image reported a more positive learning experience, but 35 percent of students chose not to view the instructor image, in part because they thought it would be distracting. In a follow-up experiment, students learned from videos that either showed the instructor's image throughout the video (constant group) or that strategically presented the instructor's image at different time points throughout the video, in an attempt to minimize distraction (strategic group). The strategic group reported higher cognitive load and feelings of social presence, but there were no significant differences between the two groups on learning outcomes. This finding is consistent with the image principle, which shows including the instructor's image does not necessarily support learning outcomes.

Embodiment Cues

Do on-screen virtual agents support learning? According to the embodiment principle, the effectiveness of pedagogical agents may depend on the extent to which they

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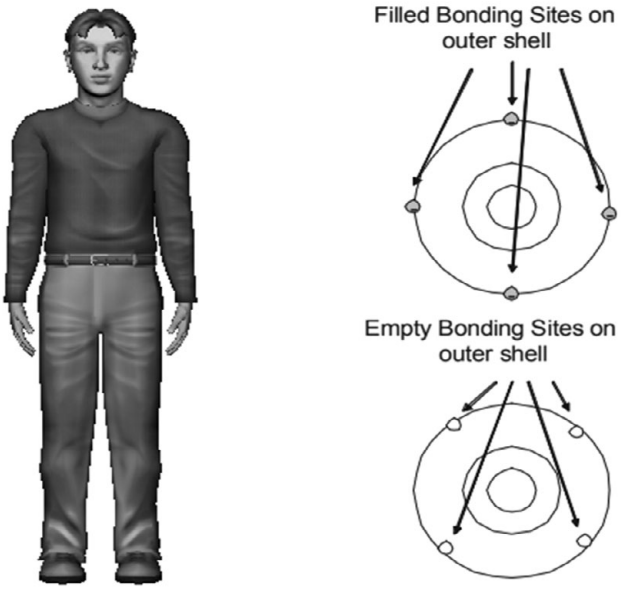

Figure 22.3 Screenshot of multimedia lesson on solar cells (from Mayer & DaPra, 2012)

exhibit human-like social cues, such as gestures and facial expressions. In a study by Mayer and DaPra (2012), college students learned from a multimedia lesson about solar cells. The lesson included an on-screen pedagogical agent who was either static (static agent group) or who displayed human-like gestures, eye movements, and facial expressions (embodied agent group). Figure 22.3 presents a screenshot from the solar cell lesson. Consistent with social agency theory, students reported the embodied agent better supported learning, was more credible, and was more engaging than the static agent. Furthermore, the embodied agent group significantly outperformed the static agent group on a subsequent transfer test. This result suggests adding specific social cues to on-screen pedagogical agents fosters student engagement and learning. The same principle is expected to apply to human instructors.

WHAT DO WE KNOW ABOUT PRINCIPLES BASED ON SOCIAL CUES IN MULTIMEDIA LEARNING?

Research on the Personalization Principle

The personalization principle is that people learn better when multimedia lessons are presented in a conversational or polite style, rather than a formal or direct style. The rationale for this principle is that personalized messages encourage students to engage in self-referential processing, in which they relate the learning material to their existing knowledge. Research suggests relatively minor changes in wording can yield a personalization effect. In a study by Mayer, Fennell, Farmer, and Campbell (2004), college students learned from a narrated animation about the human respiratory system. Personalization consisted of simply changing the word "the" throughout the lesson to "your" – that is, "the lungs" became "your lungs." Across three experiments, students who listened to the lesson refer to their own lungs significantly outperformed students who received the more formal message on a subsequent transfer test. Subsequent studies have produced similar results across different learning contexts, including Turkish students learning about the life cycle of stars (Kartal, 2010) and medical students learning anatomy terms (Ginns & Fraser, 2010).

Another form of personalization is to use polite rather than direct language. The rationale for politeness is that it creates a supportive social environment, in which students save "positive face" and avoid "negative face" (Brown & Levinson, 1987). Much of the work on politeness has involved the use of onscreen pedagogical agents. In a study by Wang, Johnson, Mayer, Rizzo, Shaw, and Collins (2008), college students learned from a computer-based virtual factory, in which they analyzed factory orders, developed production plans, and scheduled processing of jobs within the factory. Students interacted with a computer-based pedagogical agent that either provided direct feedback or suggestions (e.g., "Save the factory now") or polite feedback or suggestions (e.g., "Why don't we save our factory now?"). Students who learned from the polite pedagogical agent showed superior learning outcomes as measured by a posttest, and this effect was most pronounced for students without strong engineering background knowledge. Subsequent studies by McLaren and colleagues (McLaren, DeLeeuw, & Mayer 2011a, 2011b; McLaren, Lim, Yaron, & Koedinger, 2007) suggest the benefits of politeness may be more restricted to struggling students or those with lower prior knowledge. Apparently, these students benefit most from social support, which may motivate them to invest more effort to make sense of the learning material.

A meta-analysis by Ginns, Marin, and Marsh (2013) reported positive effects on retention (d = .30) and transfer (d = .54) for instructional messages presented in a conversational or polite style. Their analysis also suggested effects may be weaker for lessons longer than 35 minutes. Studies conducted since the meta-analysis by Ginns and colleagues have broadened the scope of the personalization effect and explored additional boundary conditions. A study by Reichelt, Kämmerer, Niegemann, and Zander (2014) found a personalization effect for German university and continuing education students on measures of retention and motivation but not transfer. In this study, students interacted with a computer-based lesson on Gestalt laws for longer than in many prior studies (25–30 minutes). A more recent study by Schrader, Reichelt, and Zander (2018) using the same materials suggests students' level of interest may serve as a moderator. The personalization effect was strongest for students with moderate levels of interest in the material, but weak for students with relatively low or high levels of interest.

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Studies by Schneider, Nebel, Pradel, and Rey (2015a, 2015b) have also extended personalization effects into new contexts. In one experiment (Schneider, Nebel, Pradel, & Rey, 2015a), students learned from a computer-based lesson on attribution theories for a 45-minute class period. Instructions were either presented in a direct style (e.g., "Now you have to work on the topic of attribution theories!") or a polite style ("In the next passage we could find out about attribution theories"). Students who learned from the polite instructions (either presented as text or audio) significantly outperformed the direct instructions condition on subsequent retention and transfer tests. The polite group also reported greater interest and feelings of assistance during learning, and lower levels of cognitive load.

In another study, Schneider, Nebel, Pradel, and Rey (2015b) extended the use of personalized language to youth slang. Secondary students learned from a computer-based lesson on photosynthesis in which the instructions were either presented in formal style (e.g., "very bad") or used slang terms (e.g., "wicked"). Results indicated students in the slang condition performed better on a subsequent transfer test, and that this effect was explained by greater feelings of familiarity. This study provides an additional way that subtle changes in the wording of instructional messages can have positive effects on learning.

Despite these promising findings, there are also situations in which personalization appears to have no effect or even negative effects on learning. One potential boundary condition is that there are cultural differences in how students respond to different language styles. Research by Brom and colleagues found no evidence for personalization effects among Czech students. In one study (Brom, Bromová, Děchtěrenko, Buchtová, & Pergel, 2014), students learned from a computer-based simulation on beer brewing over the course of two-to-three hours. There were no effects on immediate or delayed retention or transfer tests, even though the personalized group chose to spend much more time with the simulation. In a subsequent set of five experiments (Brom, Hannemann, Stárková, Bromová, & Děchtěrenko, 2017), students learned from a lesson on lightning formation used in previous research with American students. There was no benefit of conversational instructions. Interestingly, the students reported a preference for learning in a more direct rather than conversational style. These findings suggest cultural differences may moderate the personalization effect, though further work directly comparing students from different background is needed.

Finally, there may be some situations in which personalized instructional messages lead to worse learning. Research by Zander and colleagues (Kühl & Zander, 2017; Zander, Wetzel, Kühl, & Bertel 2017) has reported an "inverted personalization effect" when students learn about emotionally aversive topics. In these studies, students learned about cerebral hemorrhages from either formal or personalized materials (i.e., changing "the" to "you"). Eye-tracking analyses indicated students in the personalized group fixated less on the images from the lesson, resulting in worse performance on a subsequent transfer test. This suggests that, when learning materials associated with negative emotions, using depersonalized language might help distance learners from uncomfortable content.

Research on the Voice Principle

The voice principle is that people learn better when multimedia lessons are presented in a human voice rather than computer-generated voice. The rationale for the voice principle is that, even when the words in the narration are unchanged, subtle speech cues can affect whether the narrator is perceived as a human social partner or interferes with the listener's ability to comprehend the message.

Compared to other multimedia principles, evidence for the voice principle is limited. As described above, Mayer, Sobko, and Mautone (2003) found evidence for an advantage of a human voice presented in native accent for students learning about lightning formation. Similar results were reported in two experiments by Atkinson, Mayer, and Merrill (2005). High school and college students learned from worked examples on proportional reasoning presented on the computer. A narrator explained the solutions steps aloud using either a recorded human voice or machine voice (using text-to-speech software by Microsoft). Consistent with social agency theory, students reporting liking the human voice more and performed better on subsequent transfer tests. Further evidence for a voice effect was found by Mayer and DaPra (2012), in which college students learned better from a lesson on solar cells when the animated pedagogical agent used a human voice compared to a machine voice.

The evidence described above suggests using machine synthesized voices in multimedia lessons may negatively impact learning. However, recent advances in text-to-speech technology now allow designers to produce narrations that sound much like human speakers. A recent study by Craig and Schroeder (2017) tested whether using modern speech technology eliminates the voice effect. Students learned from a narrated animated lesson on lightning formation that included an on-screen virtual human. The narration was presented using either a classic or modern text-to-speech engine, or a human voice. Students who learned from the modern machine voice exhibited better transfer and learning efficiency than students who learned from the classic machine voice and the human voice. Students also perceived the modern machine voices as similarly credible and conducive to learning as the human voice. These results suggest modern voice technology eliminates and even reverses the original voice effect.

Although machine voices can now match the quality of a human voice, the accent of the voice (machine or human)

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may affect learning outcomes. In a recent study by Chan, Lyons, Kon, Stine, Manley, and Crossley (2020), college students who were native speakers of American English learned from a narrated slideshow on lightning formation. The narration was either presented in Standard American English by an American narrator without a distinctive accent, or by a narrator from Hong Kong whose native language is Cantonese and who speaks English with a distinctive foreign accent. Furthermore, some students viewed on-screen text that was identical to or summarized the narration, whereas other students did not view on-screen text. When no on-screen text was presented, students who listened to the foreign accent performed significantly worse on a subsequent transfer test than students who listened to the standard accent. However, this voice effect did not hold when students viewed on-screen text, suggesting it may help students compensate for the effects of the speaker's accent. Regardless of the presence or absence of on-screen text, students reported the foreign accent as more difficult to understand. Overall, a foreign accent can potentially interfere with making sense of multimedia messages, presumably because students have less experience decoding words presented in an unfamiliar accent. A potential remedy is to provide students with on-screen text that matches the narration, though one should also consider the possible drawbacks of redundant text (see Chapters 14 and 16).

Research on the Image Principle

The personalization and voice principles described in the previous two sections focus on how social cues enhance the instructor's *words*. The image principle concerns whether students benefit from *visual* social cues, such as viewing the image of a human instructor or a computer-based pedagogical agent. The rationale for including the instructor's image is that it may help establish a sense of partnership between the instructor and the learner. Consequently, students should be more engaged and motivated to make sense of the learning material. The rationale for excluding the instructor's image is that it may be distracting and create extraneous cognitive load. If students are attending to the instructor's image, they may allocate few cognitive resources to the essential learning material.

Prior research suggests students do not necessarily learn better when an image of the instructor or virtual agent is shown on the screen. Early work focused on the visibility of on-screen, cartoon-like pedagogical agents. In a series of experiments by Moreno, Mayer, Spires, and Lester (2001), students learned about environmental science from a computer-based simulation game. In one version, students viewed a pedagogical agent named Herman-the-Bug as it provided instructions; in another version, students received the same instructions but without the agent visible. Results indicated no significant differences between the two groups on subsequent retention and transfer tests.

Other studies with different learning materials have yielded similar results. Craig, Gholson, and Driscoll (2002) found no significant benefit of including the image of a pedagogical agent during an animated narration about lightning formation. Research by Atkinson and colleagues (Atkinson, 2002; Lusk & Atkinson, 2007) explored possible image effects when learning from mathematics worked examples. Results generally indicated that merely showing the image of the agent on the screen did not significantly boost learning. Finally, no significant image effects were found in studies involving multimedia lessons on electric motors (Mayer, Dow, and Mayer, 2003), visual perception (Domagk, 2010), astronomy (Frechette & Moreno, 2010), or solar cells (Mayer & DaPra, 2012).

The image principle has also been tested in the context of learning from video lectures (see Chapter 41). As the use of video for online instruction continues to accelerate, an important practical consideration is whether viewing the instructor's face fosters higher engagement and better learning outcomes. Alternatively, the instructor's face may create extraneous processing by distracting students from the learning material. Much like research involving pedagogical agents, there is not strong evidence that students benefit from including the instructor's image in video lessons. For example, several studies examining the design of video modeling examples suggests that while students spend a considerable amount of time fixating on the instructor's face when it is visible, this has a minimal impact on learning outcomes (e.g., van Gog, Verveer, & Verveer, 2014; van Wermeskerken, Grimmius, & van Gog, 2018; van Wermeskerken, Ravensbergen, & van Gog, 2018). Other studies using different learning materials report similar findings (Colliot & Jamet, 2018; Fiorella, Stull, Kuhlmann, & Mayer, 2019; Wang & Antonenko, 2017). In one study by Wilson, Martinez, Mills, D'Mello, Smilek, and Risko (2018), the instructor's presence impaired comprehension, yet students reported a strong preference for this format. This suggests students' perceptions of lesson formats do not necessarily correspond with what is best for learning outcomes.

Taken together, the available research evidence suggests including the instructor's image is not likely to significantly impact learning outcomes. It is possible that viewing the instructor carries social or motivational benefits not fully captured by prior research. Further work is needed to test the long-term impact of instructor presence on student motivation and learning outcomes within authentic learning environments.

Research on the Embodiment Principle

As described in the previous section, the mere visibility of the speaker agent does not necessarily affect learning from multimedia messages. However, research on the embodiment principle suggests this may depend on whether the instructor or on-screen agent engages in specific behaviors that guide cognitive processing and create

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feelings of social presence (see also Chapter 23). As demonstrated by the study of Mayer and DaPra (2012), pedagogical agents can be more effective when they display human-like behaviors, such as eye-contact, locomotion, gesture, or facial expressions. Similarly, Li, Kizilcec, Bailenson, and Ju (2016) found that virtual agents can be as effective as human instructors, but only if they are designed to exhibit appropriate social cues.

Other research reports similar findings, although the benefits may be subtle. In a study by Lusk and Atkinson (2007), students learned from static or animated worked examples on proportional reasoning. Students listened to spoken explanations of the examples presented by an embodied pedagogical agent (who showed gesture and eye gaze cues), a static pedagogical agent, or without an on-screen agent. Results indicated only the embodied agent condition outperformed the no-agent control condition on subsequent near and far transfer tests. However, there were no significant differences in transfer performance between the embodied and static agents (although raw scores favored the embodied agent). A subsequent study by Dunsworth and Atkinson (2010) found that embodied agents were more effective than a no-agent control, yet this study did not include a static-agent control group. Overall, these studies provide partial support for the idea that embodied agents support learning.

Other research has looked more closely at specific behaviors of pedagogical agents, such as gestures and facial expressions. For example, Baylor and Kim (2009) found that students learned how to use a software program better when gestures were present, but there was no effect of facial expression. In contrast, when instruction targeted changing students' beliefs about intellectual property, viewing the agent's facial expressions (but not gestures) was more effective. This study suggests specific embodiment cues may best support specific instructional goals, such as gestures for learning procedures and facial expressions for changing attitudes or beliefs. However, not all studies find effects of displaying an agent's facial expressions or gestures. In a study by Frechette and Moreno (2010), college students learned about astronomy from no agent, a static agent, or an embodied agent that showed facial expressions and/or gestures. Surprisingly, the only significant difference across groups indicated the students learned better with the static agent than with the embodied agent displaying facial expressions.

In some studies, agents engage in generic gestures intended primarily to create a more human-like or natural experience. However, gestures may be more effective if they serve a more specific purpose, such as directing students' attention to critical visual information. In a study by Moreno, Reislein, and Ozogul (2010), students learned from a multimedia lesson on electric circuits. For some students, the lesson contained animated arrows intended to guide attention; for other students, a pedagogical agent used hand gestures to guide students' attention. Results indicated students who viewed an agent point to important information performed better on a post-test. Subsequent studies have reported similar benefits of using embodied agents to cue important information (Johnson, Ozogul, & Reisslein, 2015; Yung & Paas, 2015), though the benefits may depend on students' level of prior knowledge. In a study by Johnson, Ozogul, and Reisslein (2015), middle school students learned about electric circuits from a lesson with or without an on-screen agent present and with or without visual signals (gestures or dynamic arrows). Results were consistent with the expertise reversal effect (see Chapter 13): students with low prior knowledge benefited most from an on-screen agent that used gestures, whereas students with high prior knowledge learned better without an agent present.

Recent studies have used eye-tracking to better understand how embodied agents affect cognitive processing during learning. In a series of experiments by Wang, Li, Mayer, and Liu (2018), college students learned about synaptic transmission from a gesturing agent, a static agent, or no agent. Students who viewed the gesturing agent fixated more on the key information from the lesson, resulting in better retention and transfer test performance. A follow-up study by Li, Wang, Mayer, and Liu (2019) looked more closely at the nature of the agent's gestures. Students either viewed specific pointing gestures, generic pointing gestures, non-pointing gestures (e.g., moving hands up and down), or no gestures. Results indicated students who learned from an agent making specific pointing gestures paid more attention to the critical information and outperformed all other groups on immediate and delayed retention and transfer tests. This suggests embodied agents are most effective when their gestures serve to guide specific cognitive processes.

Overall, adding human-like qualities to on-screen agents can support learning from multimedia lessons. A meta-analysis by Guo and Goh (2015) found that embodied pedagogical agents had moderate positive effects on motivation and learning outcomes, including retention and transfer test performance. The evidence reviewed in this chapter suggests embodied agents are most effective when they create feelings of social partnership while also supporting cognitive processing necessary for meaningful learning, such as by using hand gestures to direct attention to the essential material.

WHAT ARE THE IMPLICATIONS FOR COGNITIVE THEORY?

The evidence presented in this chapter suggests social cues can motivate students to engage in cognitive processing necessary for meaningful learning. This is an important advancement of prominent theories of multimedia learning, which traditionally focus on cognitive processing without a strong emphasis on the role of social and motivational processes. For example, many principles of multimedia design focus on reducing extraneous processing (e.g., coherence or signaling principles;

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see Chapter 14) – cognitive processing irrelevant to the instructional goals – and managing essential processing (e.g., modality or segmenting principles; see Chapter 19) – cognitive processing related to mentally representing the essential learning material. Yet even multimedia lessons designed to reduce extraneous processing and manage essential processing do not guarantee students will be motivated to make sense of the learning material. This chapter demonstrates the important role of social processes in maximizing learning from multimedia lessons.

Why do social cues foster deep learning? According to social agency, cues such as the use of conversational language or on-screen agents who display human-like gestures, help create a sense of partnership between the instructor and the learner. When learners perceive the instructor as working with them to help them learn, they are more likely to uphold their end of the partnership – by trying to actively make sense of the information the instructor presents. Furthermore, different types of social cues may contribute to productive cognitive processing in different ways. For example, conversational language (e.g., referring to "your lungs" rather than "the lungs") may prime students to engage in more self-referential processing – that is, by relating the learning material to themselves and their existing knowledge. Other social cues, such as specific pointing gestures provided by an on-screen pedagogical agent, may help direct students' attention to specific components of the essential learning material. In short, social cues are effective when they motivate students to engage in appropriate cognitive processes, such as directing attention to the critical material or integrating the material with one's existing knowledge.

WHAT ARE THE IMPLICATIONS FOR INSTRUCTIONAL DESIGN?

One potential downside of many multimedia learning environments – such as illustrated textbooks, animated narrations, or video lectures – is that they are impersonal. That is, students may feel anonymous or a lack of social agency because they are not physically present with a human instructor. This chapter suggests students do not need to be in a live classroom to feel a social connection during learning. Subtle changes to the way words or on-screen agents are presented can increase student motivation to engage in deep cognitive processing. For example, rather than using dry or formal language, instructional designers can make minor changes to convey a more personalized or conversational style, such as by using first- and second-person pronouns or by making polite statements. Furthermore, rather than merely displaying a virtual agent on the screen – which the image principle suggests doesn't greatly impact learning – instructional designers should consider ways on-screen agents can display more human-like behaviors, particularly those that serve to guide students' attention toward the essential learning material. Overall, leveraging the power of social cues can complement more conventional, cognitive approaches to effective multimedia design, such as those described in Parts IV and V of this handbook.

WHAT ARE THE LIMITATIONS OF PAST RESEARCH AND SOME DIRECTIONS FOR FUTURE RESEARCH?

Prior research on principles based on social cues has mostly involved short-term laboratory studies with college undergraduates. Therefore, we do not know whether the effects of social cues persist over longer time spans, such as longer lessons or over the course of several weeks completing an online course. On one hand, benefits of social cues may be ephemeral because students habituate to learning from a conversational style or from human-like virtual agents. On the other hand, benefits of social cues may be durable because they motivate students to persist through complex material, and to continue to engage with the course material over several weeks. Future research is needed to test the longitudinal effects of designing multimedia lessons that incorporate social cues.

Another direction for further research concerns measures of social processes during learning. According to social agency theory, social cues activate a social response in learners, which results in greater motivation to learn, higher effort, and better learning outcomes. Most measures of social agency in prior research involve asking students to self-report their feelings of social presence, engagement, or motivation during learning, or the likability or credibility of the instructor or virtual agent. More precise and objective process measures are needed to precisely capture how social cues affect social agency, motivation to learn, cognitive processing, and meaningful learning.

Future research is needed to specify the unique benefits and boundary conditions of learning from social cues. Although there appear to be benefits of adding social cues to multimedia lessons, we need a more complete understanding of how specific types of cues support learning in different contexts and with different learners. For example, we need a clearer set of guidelines for how to optimally design on-screen virtual agents that display appropriate embodiment cues. Similarly, we need a better understanding of what forms and what level of personalization is optimal when crafting instructional messages. Furthermore, future research should identify whether social cues are especially important for certain learners, such as those with little background knowledge or relatively low motivation, or when social cues are paired with established cognitive principles of multimedia design. Overall, although still a relatively nascent area, research on social cues has the potential to provide a more complete picture of how to foster meaningful learning from multimedia lessons.

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